

EXPERIMENT 5

SIMULATE THE LIFECYCLE STAGES FOR UI DESIGN USING THE RAD MODEL AND DEVELOP A SMALL INTERACTIVE INTERFACE

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AIM:

The aim is to demonstrate the lifecycle stages of UI design via the RAD model and develop a small interactive interface.

PROCEDURE:

Simulating UI Design Lifecycle Using the RAD Model

The RAD Model focuses on quick development, rapid prototyping, continuous testing, and frequent user feedback to improve the user interface efficiently.

Phase 1: Requirements Planning

Step 1: Identify Key UI Features

Define the main screens and functionalities required for the application:

- Home Page
- Product Categories
- Product Listings
- Product Details
- Shopping Cart
- Checkout
- Order Confirmation
- Order History

Step 2: Define User Actions

Identify the common tasks users will perform:

- Browsing products
- Searching for products
- Viewing product details
- Adding items to cart
- Completing checkout
- Tracking orders

Step 3: Prepare Requirements Document

Create a simple document including:

- Features and functionalities
- User stories and use cases
- Navigation flow between screens
- Functional and UI requirements

This phase helps developers and stakeholders clearly understand the project goals before design begins.

Phase 2: User Design

Step 1: Create a Project in Axure RP

1. Install and launch Axure RP.
2. Create a new project and provide a project name.
3. Select a suitable workspace for designing prototypes.

Step 2: Design Wireframes

Create low-fidelity wireframes for the following screens:

- Home Page
- Product Categories
- Product Listings
- Product Details
- Cart

- Checkout
- Order Confirmation
- Order History

Use widgets like:

- Buttons
- Images
- Text fields
- Navigation menus
- Product cards

Step 3: Add Interactions

1. Select UI elements such as buttons or icons.
2. Add interactions like:
 - **OnClick → Navigate to Screen**
 - Show/Hide Panels
 - Open Pop-ups
3. Define transitions between pages for better navigation flow.

Step 4: Create Reusable Components

Use **Masters** to create reusable UI elements such as:

- Headers
- Footers
- Navigation bars
- Search bars

This reduces repetitive work and maintains design consistency.

Step 5: Add Notes and Annotations

Add annotations describing:

- Screen functionality

- User interactions
- Navigation behavior
- Design explanations for developers and stakeholders

Phase 3: Construction

Step 1: Develop Interactive Prototype

Convert wireframes into a fully interactive prototype by:

- Adding clickable buttons
- Creating dynamic panels
- Adding page transitions and animations

Step 2: Test the Prototype

1. Preview the prototype using Axure's preview feature.
2. Perform usability testing with users and stakeholders.
3. Observe navigation flow and interaction issues.

Step 3: Improve the Design

1. Collect feedback from users.
2. Identify usability problems and design improvements.
3. Update layouts, interactions, and navigation accordingly.

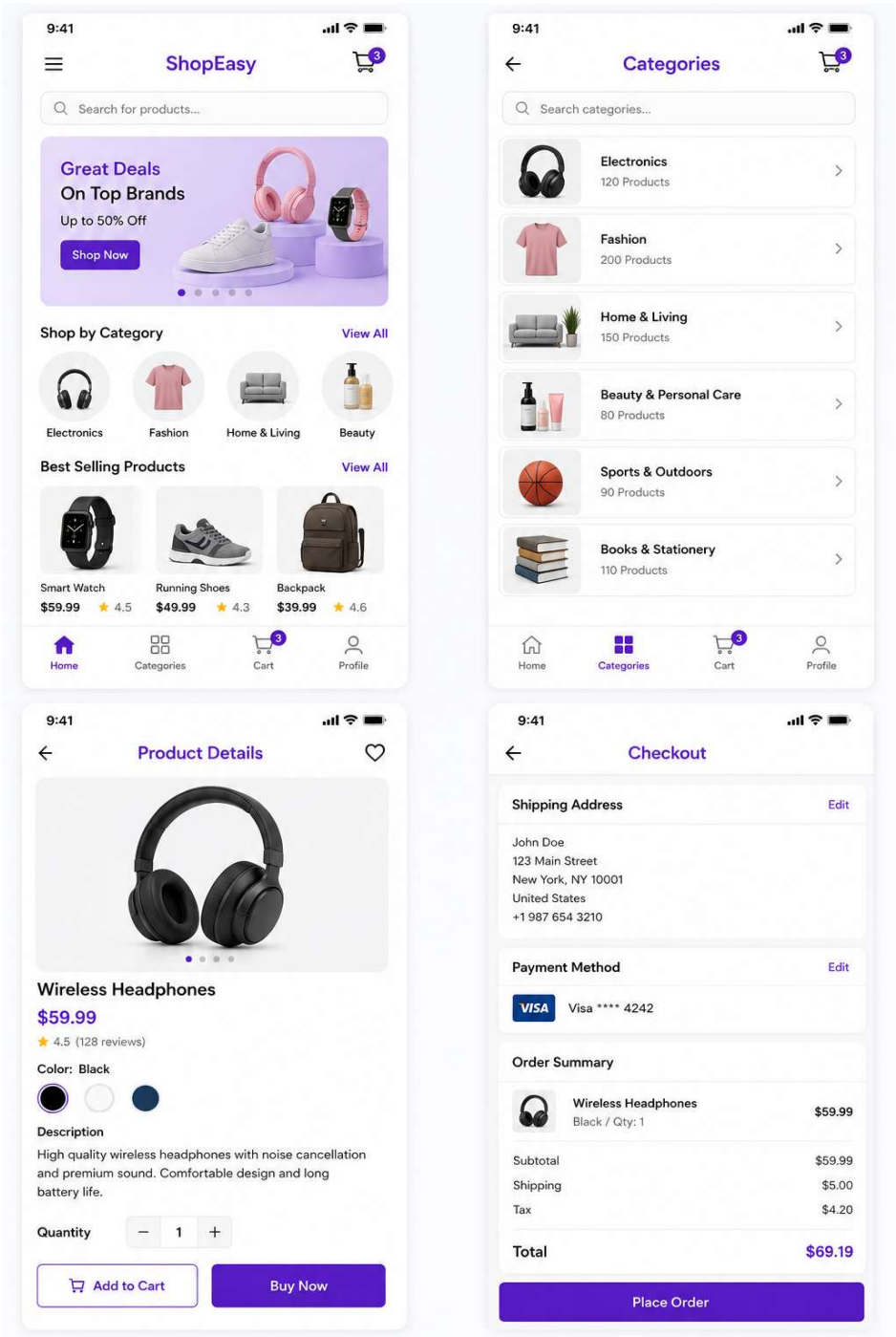
This iterative cycle continues until the UI becomes user-friendly and efficient.

Phase 4: Cutover

Finalize the Prototype

1. Review all screens and interactions.
2. Ensure consistency in design and navigation.
3. Export the prototype using:
 - Axure Cloud
 - HTML Export

OUTPUT:



RESULT

The interactive interface was successfully designed using the RAD model and provided smooth user interaction.